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Bachelor's Thesis

KAIZEN STRATEGIES FOR IMPROVING PLANNING AND PRODUCTION FLOW IN PHARMACEUTICAL ENVIRONMENTS

Double degree in Chemical Engineering and Industrial Electronics and Automation.

In colaboration with: Kaizen Institute

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Report and Annex

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Glossary

Eighty-twenty Rule Refers to the Pareto principle, which states that roughly 80% of effects come from 20% of the causes. 15

Gemba A Japanese word meaning "the real place." In a KAIZEN™ context, it usually refers to the place where value is added, such as the shop floor. It can also refer to any place where work is being performed. 9

GQCDM Principles An acronym for Growth, Quality, Cost, Delivery, and Motivation. It represents a strategic framework used in business and manufacturing to guide performance improvements.. 12, 19

JIT Just In Time. A system that "pulls" parts through production based on customer demand rather than "pushing" them based on projected demand. It results in line balancing and little to no excess inventory. 8, 9

KAIZEN™ A Japanese term meaning "change for the better." It is a gradual, long-term approach that focuses on small, incremental changes to improve efficiency and quality. It was popularized by Masaaki Imai. 6, 8–10, 12, 13, 18, 19, 22, 26

KPI An English acronym for Key Performance Indicators, which are the key indicators of a given process or activity. 10, 11, 13, 23, 26

Lean An English term meaning "no fat." In a management context, it is a strategy to reduce or eliminate waste or excess. 8, 9

Muda A Japanese word for "Waste" and a key concept in TPS. It is one of three types of deviation from optimal resource allocation (Muda, Mura, and Muri). 8, 9, 13

OPL Stands for One Page Lesson. Is a simple, visual, and concise training document used on the shop floor to quickly and effectively transfer knowledge. The key characteristic of an OPL is its brevity; it contains all the necessary information for a specific task or procedure on a single page. OPLs are typically created and maintained by the operators themselves, ensuring the content is practical and relevant. They are a core tool in Lean Manufacturing for standardizing best practices, providing quick reference for machine setup or troubleshooting, and promoting continuous improvement. They serve as a powerful alternative to lengthy manuals, making training accessible and immediate.. 16

SMED Stands for **Single-Minute Exchange of Die**. A Lean manufacturing methodology for drastically reducing the time it takes to complete equipment changeovers. The goal is to make changeovers so quick that it becomes economically viable to produce smaller batch sizes, reducing inventory and improving production flexibility.. 15, 16, 18

TPS Toyota Production System. A methodology that resulted from over 50 years of KAIZEN™ at Toyota. It is built on a foundation of Leveling with supporting pillars of Just-in-Time and Jidoka. 8

Visual Management A set of techniques for creating a workplace that uses visual communication and control. The philosophy is "what gets measured and displayed, gets done." Simple tools are used to identify the target state and correct any deviations. 9

VSA Stands for Value Stream Analysis. A Lean management method used to visually map and analyze the flow of materials and information required to deliver a product or service to a customer. VSA identifies value-added and non-value-added activities, highlights waste, and provides a basis for Kaizen (continuous improvement) efforts. It typically includes a current-state map, future-state vision, and an actionable plan to close gaps.. 14, 18

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5 Preface

5.1 Project Origin

This project is part of the curricular internship carried out at the consulting firm Kaizen Institute, which began in February 2025. Kaizen Institute specializes in continuous improvement, a concept that will be defined in the first chapter (**??**) as it provides the framework for understanding this project's approach.

The project originated when *Sincrofarm*, a pharmaceutical manufacturing company, decided to initiate a transformation process in its facility and contacted Kaizen Institute for guidance and advisory support throughout the process.

The company's production model is similar to that of a laboratory: it operates in sections, producing different elements separately. Each section has specific cycle times and production capacities that are not always compatible with those of other sections, resulting in waiting times, overproduction, or, in some cases, underproduction. Furthermore, due to stringent internal quality standards and regulatory requirements, a significant portion of operating time is spent on non-value-added activities, such as cleaning equipment or performing machine changeovers. This way of working has been effective for over forty years, but the company's exponential growth now prevents it from meeting order demand consistently, limiting its competitiveness in the sector. For this reason, modernization and adaptation to customer demand are essential to meet deadlines and increase profitability.

The project discussed below is extensive and divided into two stages: increasing line efficiency and enhancing line productivity. A daily management system called Daily Kaizen was also implemented during the course of the project; however, the details of this system are beyond the scope of the present work.

5.2 Motivation

This project is highly rewarding both personally and professionally, as it presents the challenge of completely transforming a production plant, overseeing the proposed changes, and supporting the personnel throughout the process. It provides the opportunity to drive a significant positive impact within the company—not only economically but also in terms of the work environment—by considerably improving both working conditions and ways of working.

One of the personal motivations behind this project is witnessing the practical application of our work and seeing that it has a tangible purpose. Furthermore, the project is complex: it required me to apply the knowledge gained during my university studies while also challenging my perspective, as I often had to address both technical and managerial aspects, the latter being a new experience for me. Many changes needed to be implemented, which inevitably led to unforeseen challenges along the way. This process allowed me to grow professionally, learning to tackle problems proactively, find solutions, and adapt without letting obstacles take control. Growth comes from stepping outside our comfort zone, and this project has contributed significantly to my development as a professional.

The greatest motivation, however, comes from seeing tangible results: observing the project from start to finish, with all its unforeseen challenges, is immensely rewarding. It provides a complete perspective of the process and enables the extraction of lessons and improvements not only for this project but also for future initiatives.

5.3 Planning and Project Stages

The project was delivered in the following stages:

Task / Phase	**Duration (days)**	**Start Date**	**Finish Date**
Study of Kaizen Production System	21	10/01/2025	07/02/2025
Initial Planning and Data Extraction Methodology			
Initial Planning	1	10/02/2025	10/02/2025
Factory Visit	1	17/02/2025	17/02/2025
Creation of Data Templates	3	19/02/2025	21/02/2025
Current State Analysis			
Data Recording Start	1	24/02/2025	24/02/2025
Shadowing (Jars / Liquids)	2	25/02/2025	26/02/2025
Solution Design			
Kaizen Boards	1	27/02/2025	27/02/2025
User Training	1	05/03/2025	05/03/2025
SMED Workshop	3	03/03/2025	05/03/2025
Implementation			
Daily Kaizen Kick-off	1	03/03/2025	03/03/2025
Cleaning Standard Prodcedure	2	06/03/2025	07/03/2025
General Standard of the Worker	2	14/03/2025	20/04/2025
Rollout			
Validation	31	10/03/2025	31/07/2025
Data Analysis	12	10/03/2025	31/07/2025
Control			
Steering Committee	6	17/02/2025	31/07/2025

Table 1: Summary of project tasks and durations by phase

In the Appendix A, Figure 10 a much more detailed Gantt Chart of the project phases is included for further reference.

6 Introduction

6.1 Introducing the company

Sincrofarm is a pharmaceutical company located in Catalonia, specializing in Contract Manufacturing and Packaging Services for medicines and dietary supplements, both for human and veterinary use.

Its facilities are equipped with state-of-the-art machinery and cover a total surface area of 7,200 m^2, distributed across two specialized plants: one dedicated to pharmaceutical manufacturing and the other to nutraceuticals.

Some of the main products manufactured include:

- Collagen supplements
- Magnesium citrate supplements
- Lactase-based solutions
- *Lactobacillus*-based probiotics

The company employs more than 400 people, with approximately 80

Production is organized into several manufacturing areas and work lines:

- **TMP (Raw Materials Handling)**: Responsible for preparing and mixing the active components, whether liquids or solids, with the required excipients.
- **PI (Compression)**: Dedicated to compressing solid mixtures into tablets.
- **PAC (Packaging and Conditioning)**: Responsible for product presentation and packaging, with three main formats:
  - **Liquids**: Bottled liquid products.
  - **Jars**: Solid tablets packaged in containers.
  - **Pouches**: Single-use sachets containing uncompressed solids.

6.2 Scope and pilots

The scope of this project can be narrow down into 2 *KAIZEN™ Events*, each one will be implemented in a different work line as a pilot.

Each KAIZEN™ Event will be:

- Improvement of Line Efficiency.

  Pilot: "Jars" Line

  KPI: Line OEE

  Goal: 40% OEE

- Improvement of line Productivity

  Pilot: "Liquids" Line

  KPI: Line Productivity

7 The KAIZEN Methodology

7.1 KAIZEN Origins

After World War II, Toyoda Kiichiro, then president of the Toyota Motor Company, saw the need to catch up with America to assure the survival of the automobile industry of Japan. Taiichi Ohno, Toyota's engineer, had a challenge: how to build a diversified line of products with the limited equipment Toyota possessed at the time. Instead of using batch production, he designed a system of integrated production that used one-piece flow, demonstrating that a trade-off between quality and productivity does not need to exist [1]. Aligned with other practices, Toyota Production System (TPS) was born to defy the premise that bigger batches mean lower costs.

TPS, which ultimate goal is the absolute elimination of waste, is based in two pillars: just-in-time (JIT) and autonomation (automation with a human touch). The lack of the necessary capital resources to support high inventory levels was solved with the implementation of a just-in-time pull system. JIT means "to produce the necessary units at the necessary quantities at the necessary time". A company establishing this flow throughout can approach zero inventory. With autonomation, Toyota intended to create a built-in automatic checking system against small abnormalities. With this prevention mechanism, defective products were not produced. That was not the only advantage. While the machine is working properly, employees are liberated. Consequently, one worker could attend multiple machines, increasing production efficiency. Furthermore, autonoma- tion came as a facilitator of improvement. Stopping the machine creates awareness and, through the total comprehension of the problem, improvement measures can be taken [2]. Lean, based on TPS, is a methodology that accentuates customer needs, to improve quality and reduce costs through continuous improvement and employee involvement [3]. KAIZEN™, a Japanese word, refers to the concept of continuous improvement. It does not intend to be mistaken as innovation or disruption and elevated costs. On the other hand, KAIZEN™ is about methodological improvements. It is a low-cost low-risk process that assures incremental progress and sustainable changes in the long term.

At the beginning of the 21st century, Toyota Motor Company surpassed General Motors and became the world leader in automobile production, expanding and elevating the KAIZEN™ concept for its key role in the company's success. KAIZEN™'s methodology is based on five fundamental principles described in Massaki and Sanchez's book [4] :

- Create customer value

  Value is what the customer is willing to pay for. Using a market-in approach to make informed decisions based on what the customers want and offer them that in the best way possible, improving customer experience.

- Create flow efficiency

  Flow efficiency can be obtained through the elimination of three Ms: Muda (waste), Mura (unevenness) and Muri(overburden). The concept of Muda primarily originated from Taiichi Ohno's production philosophy in the early 1950s. He defined, in the industrial context he was inserted, seven sources of Muda: overproduction, waiting, inventory, motion, transportation, over-processing and defects. All the activities that do not add value to the process are considered Muda. Fujio Cho, former President of Toyota, defined waste as "anything other than the minimum amount of equipment, materials, parts, space and worker's time, which are absolutely essential to add value to the product". Eliminating waste is the most effective way to increase productivity and decrease costs. Although Muda has gained bigger awareness, comprehending all three concepts allows for a greater understanding of Lean [1]. Mura is the variation in a process that is not caused by the final customer, for example highs and lows in the planned production due to the production system or the irregular work rhythm. Leveling production using JIT can eliminate this irregularities, avoiding work spikes and long waiting times for workers. Muri means to overburden equipment or workers, demanding a faster rhythm for a longer period than expected. While in equipment it can provoke defects and failures, in people it can result in safety issues.

- Be Gemba oriented

  Gemba is the place where value is added. Problems should be identified and solved there, at their root cause. Central to KAIZEN™ is *genchi genbutsu* (go and see for yourself). Going to the Gemba means going to the place where the action really happens, to be able to thoroughly observe the reality of the processes.

- Empower people

  Respect for people is a fundamental principle of the KAIZEN™ philosophy. Kaizen recognizes that people are not the problem, processes are. It intends to improve every day, everywhere, with everyone, training all employees and encouraging people to think and share their own improvement ideas, systematically. It is in management's biggest interest to create the best possible conditions for everyone to do the best possible work.

- Be scientific and transparent

  As important as identifying problems, is knowing how to explain and substantiate one's findings, supporting them with data. Visual Management is a basal tool of KAIZEN™, that highlights problems and allows for a quicker reaction to them.

7.2 Brief introduction to the SMED

The SMED is a technique that helps us distinguish between Internal and External tasks as described in Dillon and Shingo [5] book *A Revolution in Manufacturing*. The core principle of the SMED system is to significantly reduce the time required for machine changeovers. This is achieved by converting as many setup operations as possible from *internal* tasks (which can only be performed when the machine is stopped) to *external* tasks (which can be completed while the machine is still running). The goal is to make all changeovers last less than ten minutes, thereby maximizing production time and enabling smaller, more flexible batch sizes.

This core principle of *internal* vs *external* tasks can be applied to multiple operations, not only limited to changeovers. In this case it will be applied to Cleanings. The basics steps described in Dillon and Shingo book are as follows:

1. Identify all tasks performed during the operation.
2. Identify all participants of the operation.
3. Distinguish between *internal* and *external* tasks.

4. Convert as much tasks from *internal* to *external*.
5. Distribute and balance internal tasks in order to minimize machine stopping time.
6. Standardize procedure, create checklists and OPL.
7. Measure progress and identify bottle necks.

The SMED is a technique that helps to significantly reduce the time required for machine changeovers and setups. Developed by Shigeo Shingo in the 1950s, the core principle is to convert as many setup operations as possible from *internal* tasks (those that can only be performed when the machine is stopped) to *external* tasks (those that can be completed while the machine is still running). The ultimate goal is to complete changeovers in less than ten minutes (a "single minute" in the literal sense of the word), thereby maximizing production time and enabling smaller, more flexible batch sizes.

As detailed in Shingo's foundational book, *A Revolution in Manufacturing* [5], this methodology is not limited to product changeovers but can be applied to any process with internal and external tasks, such as the cleaning procedure identified. The SMED implementation process follows these distinct steps:

1. **Preliminary Stage**: Identify and analyze the current setup or cleaning process. This involves documenting all tasks and their timings.
2. **Stage 1: Separate Internal and External Tasks**: Critically evaluate each step of the process and categorize it as either internal or external.
3. **Stage 2: Convert Internal from External Tasks**: This is the most crucial phase. Brainstorm and implement changes that allow internal tasks to be performed while the machine is running. For example, blowing off compressed air while the line is up.
4. **Stage 3: Streamline All Remaining Tasks**: Once the conversion is complete, focus on improving the efficiency of the remaining internal and external tasks. This involves using tools like checklists, visual aids, and improved tool organization. For instance, using quick-release clamps instead of bolts.
5. **Sustain and Standardize**: Create and maintain standardized work instructions, checklists and OPLs to lock in the improvements. Establish a system for measuring and tracking progress to identify future opportunities.

This structured approach ensures a systematic and measurable reduction in changeover or cleaning time, which directly increases equipment availability.

7.3 A Brief Description on Line Balancing and Yamazumi

The foundational principle of line balancing was applied to the production process with the primary objective of distributing tasks equally among operators. This approach aimed to eliminate bottlenecks and minimize idle time, thereby optimizing workflow. The key analytical tool used for this work was the Yamazumi chart. This visual method, which literally translates to "to pile up," provided a graphical representation of each operator's work content and the time required for their individual tasks. By visualizing the cumulative workload for all team members, the Yamazumi chart immediately highlighted the imbalances and bottlenecks that were impeding the process. This visual data was critical for identifying and eliminating the forms of waste known as muda, mura, and muri—specifically, the waste of waiting, unevenness, and overburden—that were inherent in the unbalanced line. Through the strategic use of the Yamazumi chart, tasks were redistributed to create a smoother, more efficient, and balanced workflow, which directly contributed to the overall objective of maximizing productivity.

In conjunction with line balancing, the *Takt Time* is a critical metric for determining the required pace of production to meet customer demand. It serves as a heartbeat for the production process, establishing a baseline against which the balanced line's performance can be measured. The formula for calculating Takt Time is defined as (1):

$$\text{Takt Time} = \frac{\text{Total Available Production Time}}{\text{Customer Demand}} \tag{1}$$

The *Takt Time* will be used to determine the number of operators needed on the line in order to cover customer demand. As an example let's look onto the planned production for one shift. Let's say it is 26,400 units that must be covered in one shift of 7.33h. In this case following equation 1, the *Takt Time* must be:

$$\text{Takt Time} = \frac{7,33h \cdot \frac{3,600s}{1h}}{26,400units} = 0.99units/s \tag{2}$$

Meaning that the line should be averaging a production of roughly 1 unit per second.

8 Key Performance Indicators and mathematical models

8.1 OEE and profits calculation method

Overall Equipment Effectiveness (OEE) is a standard metric used to measure the efficiency of a production line. It identifies how much of the planned production time is truly productive. OEE is calculated by multiplying three core components: Availability, Performance, and Quality.

8.1.1 Availability

Availability measures the proportion of scheduled production time during which equipment is operational and ready for use. It accounts for planned downtime, such as maintenance or shift breaks, as well as unplanned losses caused by equipment breakdowns or delays.[6]

$$A = \frac{\text{Operating Time (h)}}{\text{Planned Time (h)}} \tag{3}$$

With that definition, first 2 main terms may be acknowledged:

1. Planned Time: The total scheduled production time.
2. Operating time: The Planned Time minus all planned stoppages like changeovers, cleaning, and scheduled breaks.

8.1.2 Performance

Performance evaluates whether equipment is running at its optimal speed during production. It captures losses due to reduced speed or minor stoppages that may not halt production but reduce efficiency.[6]

$$\mu = \frac{\text{Actual Units Produced (units)}}{\text{Operating Time (h)} \cdot \text{Nominal Speed (units/h)}} \tag{4}$$

8.1.3 Quality

Quality assesses the proportion of defect-free products produced relative to the total output. This metric highlights losses due to rework or defective items, which reduce the overall effectiveness of the production process.[6]

$$Q = \frac{\text{Total OK unit output (units)}}{\text{Actual Units Produced (units)}} \tag{5}$$

8.1.4 OEE calculation

By combining these three components, the final OEE formula is derived, which shows the overall efficiency of the production line.

$$OEE = A \cdot \mu \cdot Q \tag{6}$$

An OEE score of 100% represents perfect equipment effectiveness, with no downtime, optimal performance, and zero defects.[7]

Or in fair terms:

$$
\begin{aligned}
OEE = \frac{\cancel{\text{Operating Time}}\ \text{(h)}}{\text{Planned Time (h)}} \cdot \frac{\text{Actual } \cancel{\text{Units Produced}}\ \text{(units)}}{\cancel{\text{Operating Time}}\ \text{(h)} \cdot \text{Nominal Speed (units/h)}} \cdot \\
\cdot \frac{\text{Total OK unit output (units)}}{\text{Actual } \cancel{\text{Units Produced}}\ \text{(units)}} = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \cdot \text{Nominal Speed (units/h)}}
\end{aligned}
\tag{7}
$$

$$
OEE = A \cdot \mu \cdot Q = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \cdot \text{Nominal Speed (units/h)}} \tag{8}
$$

8.1.5 Profit Calculation

Now that the concept of OEE is clear, going back to equation 11. In KAIZEN™ philosophy the focus is on the non investment improvements. Changing the Nominal Speed of a line or machine could mean investing a huge amount of money in technology, instead of focusing on eliminating Muda. That's why the project focuses on the OEE as a KPI as the main metric.

We can quantify the augmented capacity derived from an improvement in OEE as the reduced planned operating time of the line. In this work we will not be focused on improving sales directly, so the Total Number of units produced will be fixed. We will not focus on investing in new equipment, so the Nominal Speed of the line will stay constant. Every point of improvement of OEE will be inversely related to a decreased in the need for more planned time which in turns decreases the operating cost of labor of the line. Relating back to the objective function, equation 10.

8.2 Productivity and profits calculation method

8.2.1 Productivity

For this study, productivity will be calculated using Equation 12. Because products produced on the same line may exhibit different standard line speeds, this factor is incorporated to enable fair comparisons across different product types. For the remainder of the analysis, we refer to productivity improvements as ΔProductivity. Measured relative to the previous standard as described in Townsend's book *The basics of line balancing and JIT kitting* [8].

$$
\Delta\text{Productivity}(\%) = \frac{\left(\dfrac{\text{Actual Unit Output}}{\text{Planned Operation Time}_1 \cdot \text{Actual N}_{\text{operators},1}}\right)}{\left(\dfrac{\text{Planned Unit Output}}{\text{Planned Operation Time}_0 \cdot \text{Planned N}_{\text{operators},0}}\right)} - 1 \tag{9}
$$

Here, the subscript 0 refers to the baseline or historical production values, while subscript 1 corresponds to the post line-balancing measurements.

8.2.2 Profit Calculation

In alignment with OEE profit analysis, each incremental gain in productivity is translated directly into cost savings by reducing the total operation time required to produce the same number of units. This approach establishes a direct quantitative link between productivity improvements and financial performance, consistent with Lean Kaizen principles of continuous improvement and cost efficiency.

9 Improvement of Line Efficiency

9.1 Objective Function

As in any optimization problem, we must first define the objective function. In the KAIZEN™ methodology, improvement efforts are directed toward what is most valuable for the client, often summarized by the principles of GQCDM Principles:

- **Growth**: Focus on expanding market share, innovation, and business development.
- **Quality**: Ensuring high standards in products, processes, and customer satisfaction.
- **Cost**: Controlling expenses and improving efficiency to remain competitive.
- **Delivery**: Meeting deadlines and ensuring timely, reliable delivery of products or services.
- **Motivation**: Encouraging and empowering employees to foster engagement, productivity, and innovation.

In this particular KAIZEN™ event, the focus will be on minimizing **Production Cost**. Specifically, this is pursued by maximizing the sustainable production capacity of the line.

$$\text{Objective function} = \max\left(\frac{\text{Total output units}}{\text{Time window}}\right) \tag{10}$$

In other words, equation 10 seeks to maximize the throughput rate, for example the number of orders the factory can process in a year.

This can also be expressed as:

$$\text{Objective function} = \max\left(\text{Rated line speed} \cdot OEE\right) \tag{11}$$

Here, the rated line speed represents the theoretical design capacity, while OEE adjusts for availability, performance, and quality losses. The introduction of this efficiency factor will be discussed in the following section.

It is evident that production output is strongly embedded in the cost structure of a company. A significant share of manufacturing cost is related to the utilization of machines and operators. By producing more in less time, the cost per unit decreases, thereby improving overall competitiveness.

9.2 Identifying improvement opportunities

A comprehensive analysis of Overall Equipment Effectiveness (OEE) provides a structured approach to identifying production inefficiencies. Following a VSA, our project focused on the key opportunities identified to improve OEE. The project's baseline OEE was established after one month of observation in February, with the following results:

- Availability=41.6%
- Performance=66.2%
- Quality=99.89%
- OEE=27.5%

The baseline data clearly indicates that low availability is the primary factor limiting overall equipment effectiveness. This is further explained in Figure 1, which shows that only 40% of the available time is utilized for production.

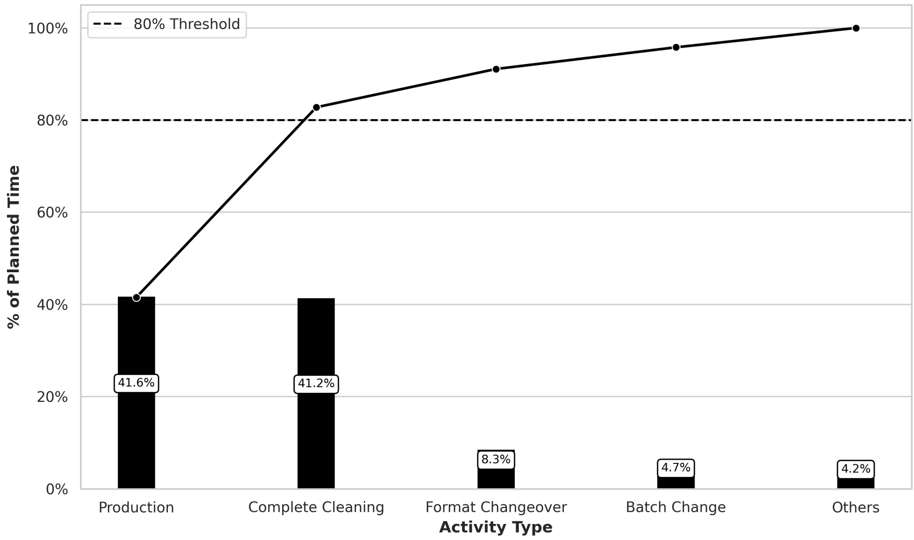

Figure 1: Pareto Chart of Planned Time dedication by Activity

The main reason for this loss in availability is the time dedicated to cleaning all machine equipment and the production space. This extensive cleaning is a direct consequence of stringent quality controls within the pharmaceutical industry, which mandate:

- The prevention of cross-contamination between different products or batches.
- The removal of product from machinery overnight to prevent quality degradation.

Figure 2, provides a more detailed breakdown of the availability losses. By applying Eighty-twenty Rule we can focus on the key areas for improvement.

The analysis confirms that a significant efficiency improvement can be achieved by reducing the time spent on cleaning. This objective will be addressed using the SMED Methodology.

9.3 Implementing the SMED Methodology

9.4 Documentation of Tasks and Timings

The first preliminary step of the SMED methodology is to conduct a thorough analysis of the existing changeover process to establish a baseline. This was achieved through direct observation and detailed time studies, utilizing video analysis to capture every task performed by the operators and the shift leader inputs to calculate their precise timings. This meticulous documentation serves to eliminate assumptions and reveal previously unnoticed process inefficiencies. A summary of the identified tasks is presented in Table 2, while the complete, detailed list can be found in the Appendix B.

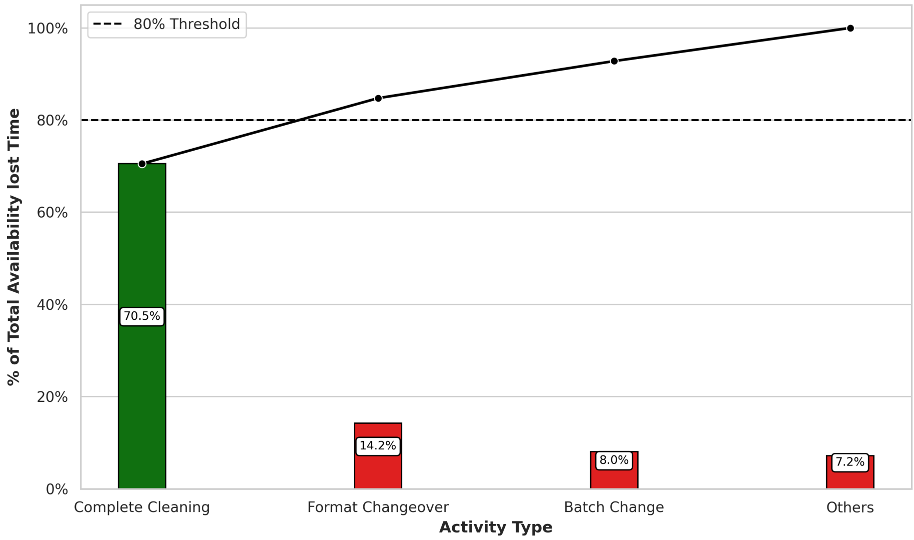

Figure 2: Pareto Chart of Main Losses on Availability. In Green Above 80% of cumulated time, in Red the rest 20% of time

Table 2: Cleaning Tasks Sorted by Duration (Descending)

Task	**Time (min)**	**Task**	**Time (min)**
Clean technical area	120	Clean secondary equipment	60
Clean ceilings and walls (secondary)	45	Clean positioner	45
Clean secondary floor	30	Clean Cremer 1	30
Clean ceilings and walls	30	Remove leftover material	25
Clean T-Delta	23	Clean Cremer 2	23
Dismantle Cremer suction equipment	15	Take carts to washing area	15
Soap cart 1 parts	15	Clean floor	15
Clear production line	13	Vacuum and blow	12
Apply disinfectant cart 1	10	Dismantle positioner	10
Dismantle Cremer 1	10	Dismantle T-Delta	10
Count materials	10	Bag and label clean parts	8

9.5 Separating Internal from External Tasks

With a comprehensive understanding of the current state, the team's focus shifted to categorizing each task. Through collaborative workshops and direct consultation with the operators we classified each task as either *internal* or *external*. This classification was visually represented in an *E-I* chart, which laid the foundation for the subsequent conversion efforts. The resulting categorization is presented in Table 3.

This analysis revealed a significant opportunity for improvement. The total calculated internal time needed was a maximum of **3h 18min**, within a total cleaning time baseline of **5h 5min**. This gap represented the potential for substantial time savings. A visual representation of all operations and their timings is presented in the Appendix B, Figure 11.

Table 3: SMED Tasks Classified as Internal and External

Internal Task	Time (min)	External Task	Time (min)
Clean secondary equipment	60	Clean technical area	120
Clean positioner	45	Clean ceilings and walls (secondary)	45
Clean Cremer 1	30	Clean secondary floor	30
Clean ceilings and walls (primary)	30	Remove leftover material	25
Clean T-Delta	23	Clean floor	15
Clean Cremer 2	23	Vacuum and blow	12
Dismantle Cremer suction equipment	15	Count materials	10
Take carts to washing area	15	Bag and label clean parts	8
Soap cart 1 parts	15		
Clear production line	13		
Apply disinfectant cart 1	10		
Dismantle positioner	10		
Dismantle Cremer 1	10		
Dismantle T-Delta	10		

9.6 Standardizing the New Procedure

To lock in the improvements and ensure their sustainability, a new standard operating procedure was developed. This was not merely a document but a comprehensive set of visual tools created in close collaboration with the operators. Every step was detailed in a clear checklist to provide a quick visual reference for the entire process.

The new procedure was treated as a living document. Any deviation from the objective time was immediately treated as a problem to be solved. This triggered a deviation where the shift leader and team would analyze the root cause of the delay, identify a countermeasure, and update the standard procedure to prevent the issue from reoccurring. This feedback loop ensured continuous improvement and maintained the gains achieved through the SMED effort.

Also the new checklist of operation added the differentiation between internal and external task, to make sure the machine availability would be hindered as less as possible form the cleaning procedure.

An example of the checklist used can be found in the Appendix B on Figure 12.

9.7 Results and Process Evolution

The baseline data, collected during my observations in February 2025, established the average time for the Jars line cleaning procedure at **5 hours and 5 minutes**. This total time was composed of **3 hours and 18 minutes** of internal operations (performed with the line stopped) and **1 hour and 47 minutes** of external operations that were also conducted during downtime.

Based on this analysis, a two-pronged strategic goal was established. The first and most immediate objective was to convert the **1 hour and 47 minutes** of external operations, which were previously performed with the machine off, to be conducted in parallel with production or during the preparation phase. This would immediately reduce machine downtime from 5 hours and 5 minutes to a maximum of 3 hours and 18 minutes, matching the internal operation time. The second objective, informed by the VSA analysis, was to systematically reduce the remaining internal time by a third, setting an ambitious final target of **2 hours and 15 minutes** for the entire cleaning procedure.

The evolution of the cleaning process time is visually documented in Figure 3, which tracks the procedure's duration over several weeks.

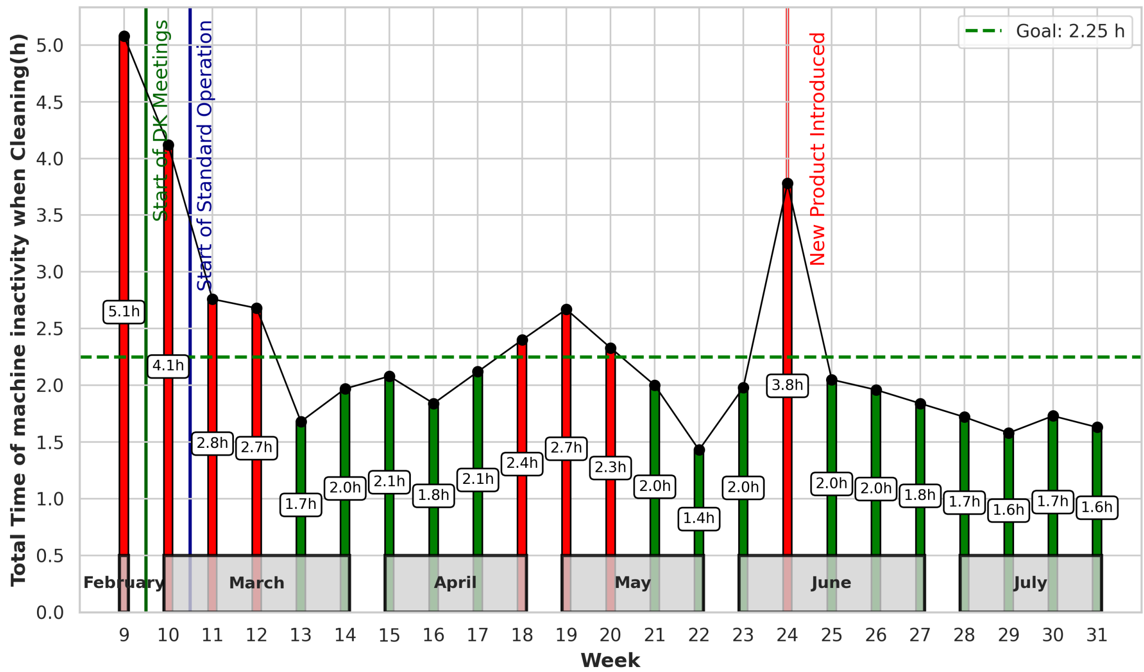

Figure 3: Evolution of the Cleaning Time Procedure along the Weeks.

The data reveals a clear and immediate reduction in downtime following the introduction of the new Standard Operating Procedure and the strategic conversion of external tasks. Machine inactivity levels dropped drastically, reflecting the direct impact of the SMED methodology. The figure also demonstrates a clear downward trend in cleaning time over the subsequent months, indicating a culture of continuous improvement and effective knowledge sharing among the operators. This sustained improvement is a powerful testament to the successful application of KAIZEN™ principles beyond the initial project scope.

A good example of this came in week 24, when we introduced a new, new, high-viscosity product tha disrupted established processes. At first, the adhesive characteristics proposed an unexpected challenge threatening the steady progress on the adaptation of the new standard. However, the team quickly used the standard procedure and checklist as a guide to break the problem down and adjust on the fly. Operators worked together to test ideas, tweak tools and cleaning products. Settling on a better method. By the next week, they had updated the standard and were back on track with our time targets. This showed just how adaptable the team had become and how a strong Kaizen culture helps us solve problems quickly and lock in improvements.

As a conclusion, from the first observation of a Cleaning machine downtime of **5h 5min** to the final observation right before summer break of **1h 36min** there is an improvement of **68%** reduction on machine downtime during Cleaning procedures.

10 Improving Line Productivity

10.1 Objective Function

This KAIZEN™ event is grounded in GQCDM Principles principles, with a primary focus on enhancing line productivity through the optimal allocation of human resources necessary to complete a production batch. The analytical perspective shifts from traditional machine efficiency metrics to the evaluation of human time investment, identifying labor allocation as a critical lever for productivity improvement.

The objective function for this Kaizen event is formally defined as the maximization of productivity, as expressed by the following equation:

$$\text{Objective Function} = \max\left(\text{Productivity}\right) = \max\left(\frac{\text{Unit output (units)}}{T_{\text{operation}},\,(\text{h}) \cdot N_{\text{operators}}}\right) \tag{12}$$

Where:

- *Unit output*: Represents the total number of units produced within a defined production batch.
- $T_{operation}$: Denotes the total person-hours invested in production, expressed as the sum of all operator working hours.
- $N_{operators}$: Refers to the number of operators assigned to the production line.

Maximizing productivity, as defined by this function, inherently targets production cost reduction. By producing the greatest unit output for the minimum operator time, the event directly aligns with Kaizen principles of waste elimination and resource optimization, as each operator-hour represents a significant and reducible cost.

10.2 Initial Line distribution

As stated in previous sections, the pilot line for this endeavour was the "Liquids" Line of the plant. In this line drugs sold and administered in liquid form are bottled, boxed, added with the dosification device necessary, packed into distribution boxes and palletized ready for shipment.

It is divided onto 2 main section, the primary section, and the secondary. In the pharmaceutical industry, the primary section is where the product is treated outside of its shipping container, in this case its where the liquid medicine comes out from the reactor into the bottler. The primary section is regulated under heavier security measures as the product is in contact with the environment at it is more likely to be contaminated. The secondary section is where the conditioning of the bottle takes place, this is the labelling, boxing, prospect insertion, dosser insertion, and further transformation occurs. As the product is already bottled up and has no interaction with the environment the regulations are much looser. The two sections are physically separated by a positive pressure wall and to access the primary one must change his attire into a more restricted one which in turn difficult the movement between the two sections.

As depicted in the line schematics, one person must reside at all times in the primary section and as movement between the two section is heavily restricted the focus of this project will be put specially on the secondary section of the line, as not only it host the most amount of resources it is where the majority of the transformation takes place.

As shown in Figure 4 the line hosted 2 operators and 1 line coordinator.

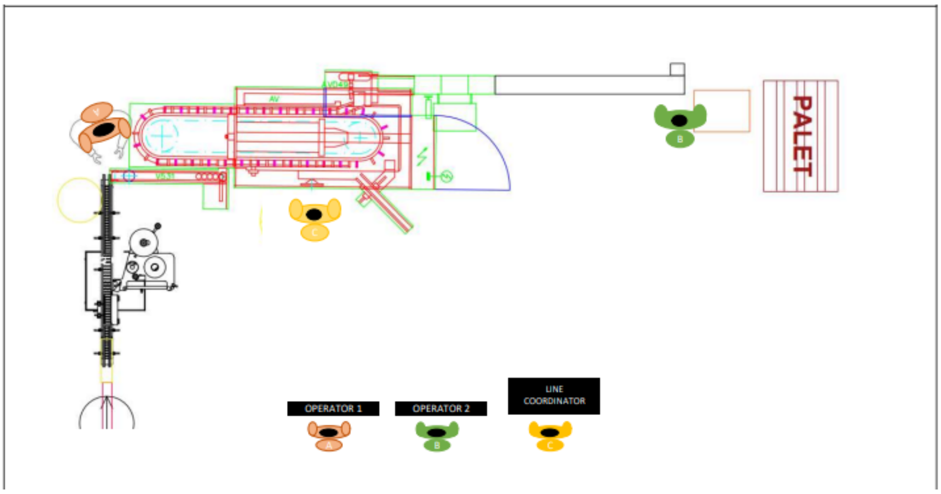

Figure 4: Line Diagram of Operator distribution

After an initial observation round. Every task done by each operator during the shift was recorded, classified and inspected. The table showing details can be found in the Appendix C Table 6.

The line *Takt Time* was calculated to 2.8s. The Yamazumi chart in figure 5 shows task distribution between operators. It is clear that operator occupation is not maximized. This chart shows us that the need for 3 operators in the line is not due to high occupation and line tasks but rather due to a lack of time organization and task sequencing.

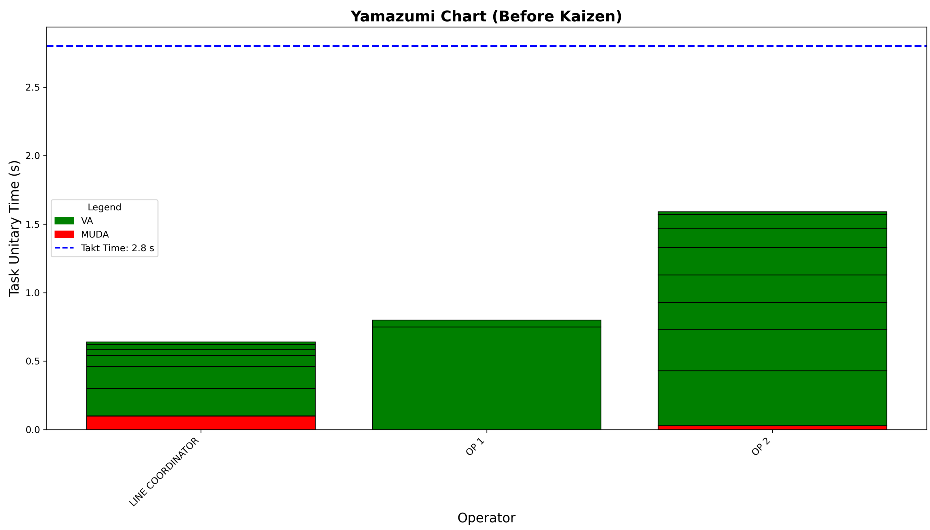

Figure 5: Yamazumi Chart Showing Task distribution before improvement.

10.2.1 Line Balancing

The main problem was indeed time organization. Some operations had to be performed simultaneously. Like quality checks, reloading leaflets or cases. Changing the optics from the operations, it was noticed that the bottle neck of the line did not come from the secondary section but rather how fast the primary bottling would go. So it would be not a problem to speed up the secondary cycle time and have some slack to stop the line for a few minutes in order to conduct some of this tasks, like quality controls, loading of leaflets, changing labels, change cases, etc. Meanwhile build up a buffer in the line and systematically clear it up. Maintaining this way the average line *Takt Time*.

Organizing the time in this matter and reducing the Non-added Value activities as much as possible, the line tasks where reorganized to fit only 2 operators.

The new Yamazumi chart with only 2 operators is shown in Figure 6.

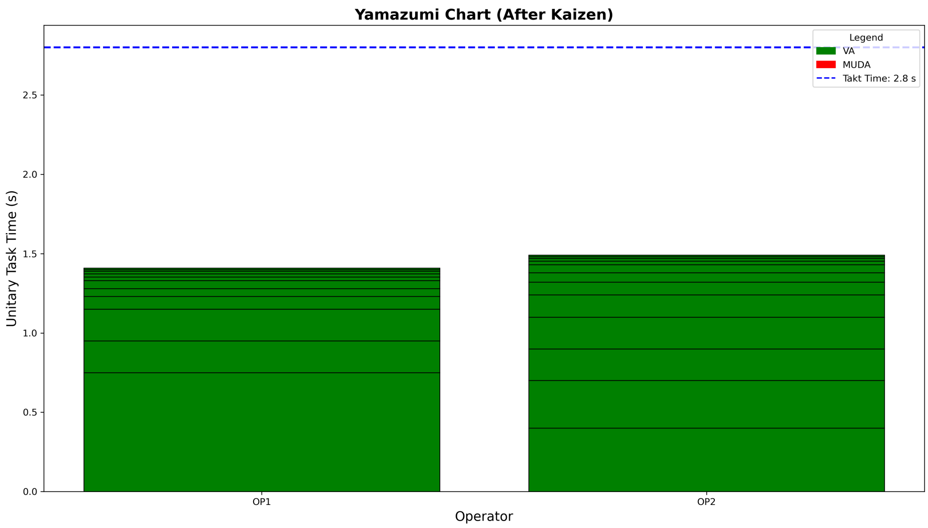

Figure 6: Yamazumi Chart Showing Task distribution with 2 operators.

In the end, the deliverable of this KAIZEN™ Event is the generation of the General Standard of the Worker, which describes how the line should be operated with the planned resources. This Standard can be found in Appendix **??**

10.3 Results and Evaluation

Coming back to equation 9 where we described the method for calculating the increment in productivity, here, as line *Takt Time* and Unit Output stay the same we can simply calculate the theoretical gain on $\Delta Productivity$ as:

$$\Delta Productivity(\%) = 1 - \frac{N_{operators,1}}{N_{operators,0}} = 1 - \frac{2}{3} = 33\% \tag{13}$$

Meaning a direct increase in productivity expected should be of **33%**.

However in practice a reduced line efficiency was observed. The operators could not keep up with the tasks at hand. This placed the challenge to modify the current line pace. After careful analysis it was concluded that the bottle neck for the line came from the bottling rate in the primary section. Which meant that the main secondary machine, the leaflet placer could be stopped at times and then sped up systematically in order to catch up to the primary production without ever compromising the line bottle neck. This method will be address as *Buffering* from now on, as it builds up a buffer of product on the entrance to the leaflet placer and allows operators to perform routinely tasks without compromising line efficiency.

In Figure 7 the evolution of the $\Delta Productivity(\%)$ can be observed. Notice that in the first week of applying the new line operator distribution, planned production was not met. As described earlier operators could not keep up with the task demand whilst maintaining line efficiency so the *Buffering* method was introduced.

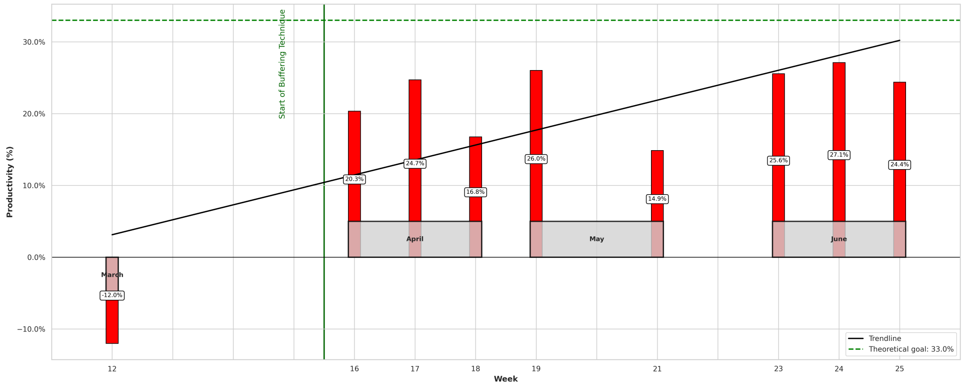

Figure 7: Evolution of $\Delta Productivity(\%)$ along the weeks

This particular line only ran 9 weeks from March until June, as it is not one of the saturated lines in the plant. However for the 8 weeks that run with the new buffering method and General Standard of the Worker, productivity increased an average of **22.47%**.

11 Results

11.1 Improvement on line efficiency

11.1.1 OEE Improvement

In the project described on section 9 the main KPI for that project was **OEE**. In Figure 3 a clear downwards trend on cleaning time was observed. In fact a total reduction of over 66% of cleaning time.

This reduction of cleaning time impacted directly to the OEE by means of the Availability metric. In the Figure 8 the evolution of the OEE within the weeks is depicted.

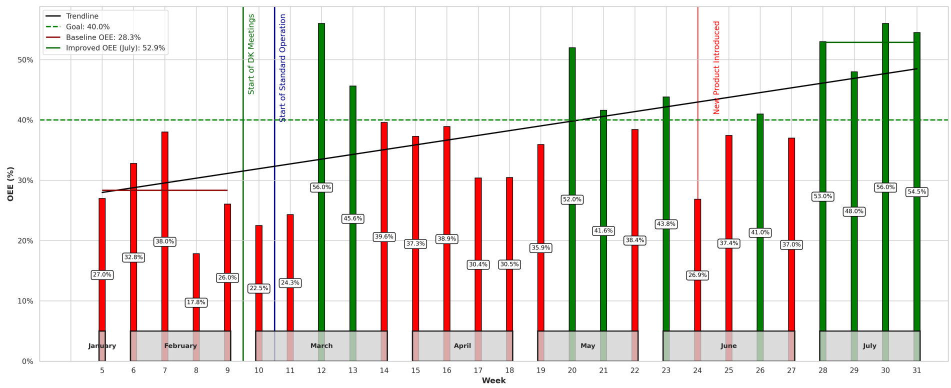

Figure 8: Evolution of OEE along the weeks

As it is shown the improved trend for the OEE is clear. From a Baseline recorded the first 5 Weeks of an average of 28.3% to an average the last month of recorded data of 52.9% a 14 point improvement in OEE metrics.

11.1.2 Production Capacity Improvement

As explained in section 8.1.5 combining Equation 10 and 11 get's us the following notation:

$$\text{Production Capacity}_0 = \frac{\text{Total Output Units}}{\text{Time Window}_0} = \text{Line Rated Speed} \cdot \text{OEE}_0 \tag{14}$$

$$\text{Production Capacity}_1 = \frac{\text{Total Output Units}}{\text{Time Window}_1} = \text{Line Rated Speed} \cdot \text{OEE}_1 \tag{15}$$

As Total Output units and Line Rated Speed remain a constant and for the profit calculation method the Time Window_0 will be the entire production year, 218 days or 3,488 available hours.

$$\text{Time Window}_1 = \frac{\text{OEE}_0 \cdot \text{Time Window}_0}{\text{OEE}_1} = \frac{28.3\% \cdot 3,488h}{52.9\%} = 1,866h \tag{16}$$

Meaning that the same production output can be achieved in almost half the time. This reduction of the 46% of the necessary production time.

The line currently employs 4 operators per shift, running 2 full shifts 5 days a week. By reducing the production time from 3,488 hours/year to 1,866 hours/year, saving 6,488 hours/year of direct operator working time.

The current hourly rate for a pharmaceutical operator in Spain is approximately 16.4€ before taxes [9].

A total savings of 6,488 operator hours a year represents a total savings amount of **106,400€/yearly** saved in direct labor work.

11.2 Improvement of line productivity

On the other hand, there is the results form the project described in section 10.

The Liquids Line is not a saturated line in the plant. Which mean the planned time for the line is only the 9 total weeks the exercise was run on and there weren't any other Work Orders placed in the line for the remanding of the year. The profit calculation approach follows a similar logic as the section before. The total hourly plan of the line yearly was of 720 h. From Equation 9 it is possible to isolate the parameter $Operation\ Time \cdot N_{\text{operators},1}$ as this are the total number of worked hours by operators.

To achieve the same production there is now a requirement to operate the line during 880 h but the total worked time by operators comes down to 1,760h versus the previous 2,160h. This savings of 400h at the same 16.4€ as previously, comes to a total saving of **6,500€/yearly**.

12 Sustainability and Ethical Analysis

The present project, responds to the industrial need for higher efficiency and competitiveness. Beyond the technical and economic outcomes, it is necessary to reflect on the environmental, social, and ethical dimensions.

12.1 Environmental Sustainability

Two dimensions were considered following the EEBE sustainability cell framework:

- **During the project execution:** The main environmental impacts were related to consultancy activities (energy consumption for IT equipment, transport to the job site, and limited use of office materials). The carbon footprint was minimized through digital documentation and reduced printing.

  The carbon footprint calculated for the transportation, traveling 40km per trip during 51 days. Using a gasoline car with a consumption of 7.3 l/km. The carbon footprint for transportation was of 303 kg_{CO2eq}

  The carbon footprint for the IT equipment used in the 660h of a laptop consuming 65W. Taking in account the average emission rate of Spanish electricity production at $157g_{CO2eq/kWh}$ [10]. The carbon footprint for IT equipment was $67,35$ kg_{CO2eq}

- **During the implementation of results:** The reduction of times leads to lower energy and water consumption on the production line. Additionally, fewer cleaning cycles result in a reduction of chemical residues, contributing to a smaller environmental footprint of the plant's operations.

Some measures to limit the total footprint of the project were to limit the visits to the job site to the strictly necessary.

12.2 Ethical Implications

The following ethical aspects were considered:

- **Needs addressed:** The work directly responds to the demand for reducing non–value-adding activities and increasing production efficiency. These needs were defined by the company's operational strategy but are also aligned with the broader objective of responsible resource management (contributing to several UN Sustainable Development Goals, particularly SDG 9 and SDG 12) [11]
- **Beneficiaries:** The company benefits in terms of efficiency, cost savings, and competitiveness. Employees may benefit from reduced workload pressures and more structured workflows. No collective or community has been identified as negatively affected by this work.
- **Societal benefit:** Increasing efficiency in pharmaceutical production lines contributes to more affordable medicines and greater access to essential health products, thereby generating a positive societal impact.

- **Unintended consequences:** Potential risks include excessive focus on labor cost savings that could be misinterpreted as justification for workforce reductions. The project therefore emphasizes the use of time savings to reinvest in higher-value tasks and continuous improvement activities.
- **Deontological aspects:** The work respects professional ethical standards, ensuring transparency, data integrity, and alignment with industrial safety and regulatory frameworks.

Figure 9: SDG (Sustainable Development Goals) by United Nations [11].

12.3 Overall Assessment

From both ethical and sustainability perspectives, the project is aligned with the principles of responsible engineering practice. It fosters efficiency without compromising workers' rights, improves resource utilization, and contributes positively to the environmental performance of pharmaceutical operations. In this way, it supports the integration of KAIZEN™ methodologies not only as a tool for economic gain but also as a driver of sustainable industrial development.

13 Economic Evaluation

After presenting the results the topic of total project cost should be addressed. All calculations for consultancy cost will be quoted using the Spanish statute for Business Consultancy regulated statute group A3GAN1 [12]. Which states an hourly wage of 16.67€/h.

The cost evaluation will take into account:

- Direct Costs:

  Total Consultancy hours: 660h

  Transportation Costs: 12€ per day worked at job site.

  Software Subscription (Office 365 Package): 657€

  Office Materials (Paper, pens, sticky notes...): 500€ fixed cost.

- Indirect Costs

  Consultant Laptop Amortization over 5 year lifetime: 19.98€/month

- Strategic Costs:

  Overhead Margin: 10% of all cost

Cost Category	**Basis / Rate**	**Amount [€]**
Direct Costs		
Consultancy Hours	660h · 16.67 €/h	11,002.2
Transportation	12 €/day · (51 days)	612
Software Subscription (Office 365)	Fixed annual license	657
Office Materials	Fixed	500
Subtotal Direct Costs		**12,771.2**
Indirect Costs		
Laptop Amortization	19.98 €/month × (6 months)	119.88
Subtotal Indirect Costs		**119.88**
Strategic Costs		
Overhead Margin	10% of (Direct + Indirect)	1,289.10
Total Project Cost		**14,180.18**

Table 4: Cost evaluation for the project

14 Conclusions

The projects described demonstrate the tangible benefits of applying KAIZEN™ manufacturing tools, in particular SMED and line balancing techniques, to pharmaceutical production lines. Both approaches were implemented with the goal of reducing non–value-added time, increasing efficiency, and improving the utilization of available resources.

The SMED initiative led to a significant reduction in cleaning and changeover times, achieving a decrease of over 66%. This improvement directly impacted the line's *OEE*, which rose from a baseline of 28.3% to 52.9%, representing an increase of 24.6 percentage points. The resulting gain in availability translated into a potential reduction of production hours from 3,488 to 1,866 per year. When expressed in labor costs, this corresponds to savings of approximately **106,400€ per year**.

In parallel, the line balancing project demonstrated that productivity improvements can be achieved even in non-saturated lines. By redistributing operator tasks and improving workflow, the total operator working hours required for equivalent output decreased by 400 hours annually. This reduction yielded direct labor cost savings of approximately **6,500€ per year**. Although monetary savings did not keep up with the expectation, the concept was proven and in the future it will be expanded to higher demand lines.

Taken together, these results validate the effectiveness of lean methodologies in a regulated pharmaceutical environment. The combination of SMED and line balancing not only improved technical performance indicators such as *OEE* and operator efficiency, but also translated into measurable economic impact. Beyond the quantified cost savings, the projects fostered a culture of continuous improvement, increased operator engagement, and built a foundation for sustaining long-term operational excellence.

Future work could explore extending these techniques to other production lines, assessing the scalability of improvements, and integrating digital tools for real-time monitoring of KPIs. Such efforts would further enhance the robustness of production systems and strengthen competitiveness in the industry.

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Appendices

A Project Details

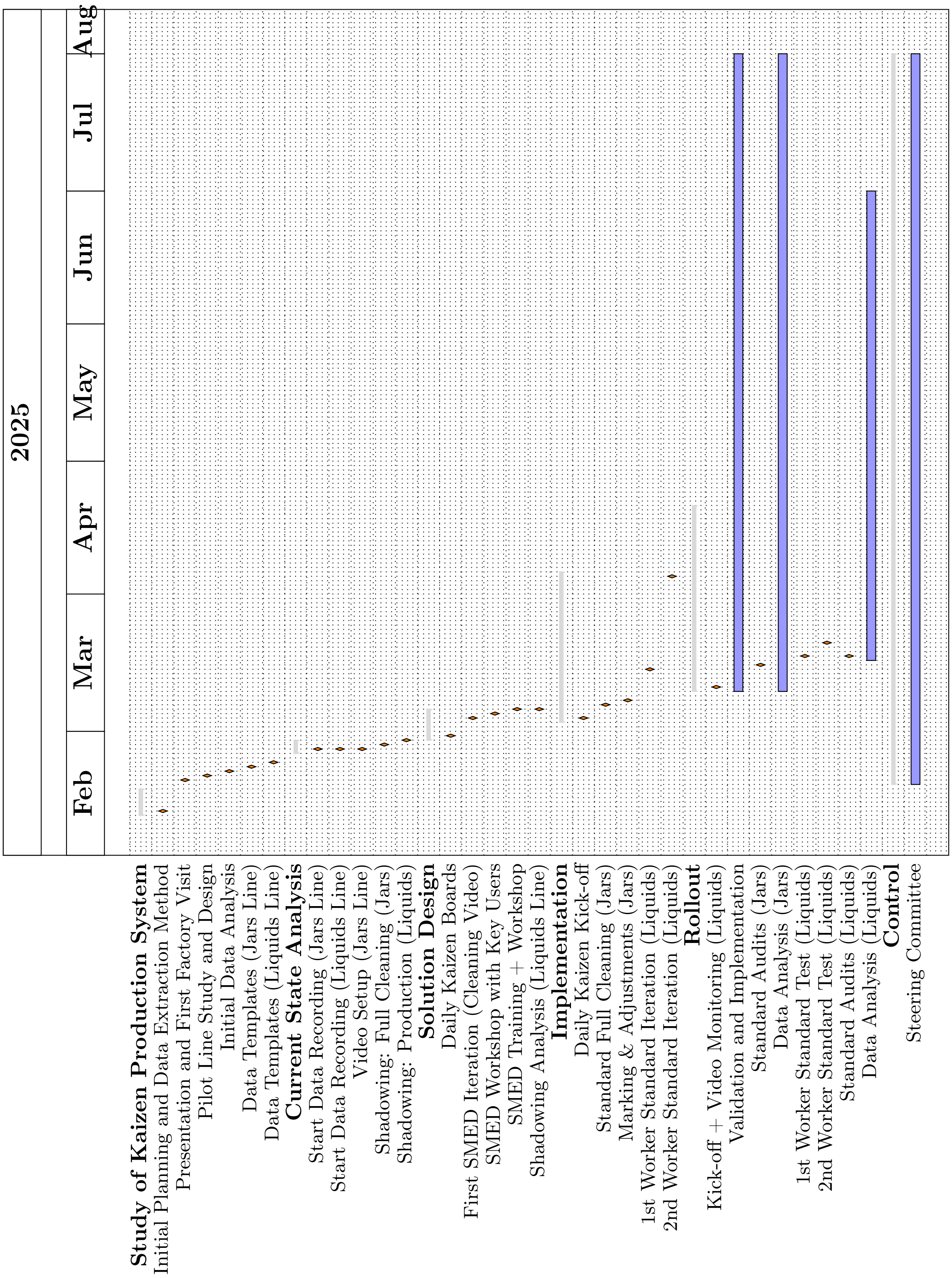

Figure 10: Detailed Project Gantt chart

B SMED Task Detail

Table 5: Cleaning and Production Tasks Sorted by Duration (Descending)

Task	Time (min)	Task	Time (min)
Clean technical area	120	Clean secondary equipment	60
Clean ceilings and walls (secondary)	45	Clean positioner	45
Clean secondary floor	30	Clean Cremer 1	30
Clean ceilings and walls	30	Remove leftover material	25
Clean T-Delta	23	Clean Cremer 2	23
Dismantle Cremer suction equipment	15	Take carts to washing area	15
Soap cart 1 parts	15	Clean floor	15
Clear production line	13	Vacuum and blow	12
Apply disinfectant cart 1	10	Dismantle positioner	10
Dismantle Cremer 1	10	Dismantle T-Delta	10
Count materials	10	Bag and label clean parts	8
Final rinse cart 1	8	Remove cleaning materials	8
Partial soaping cage 1	8	Bring cleaning materials upstairs	5
Bring parts upstairs	5	Close logs and cards	5
Access primary area	5	Initial rinse cart 1	3
Initial rinse cage 1	3	Remove materials	3
Bring carts	3	Close guide	3
Record in production log	3	Change cards	3
Remove leftover material (conditioned)	3	Bring vacuum cleaner	3
Take cart 1 to dryer	1	Insert cart 1 into washer	1

C Line Balancing Details

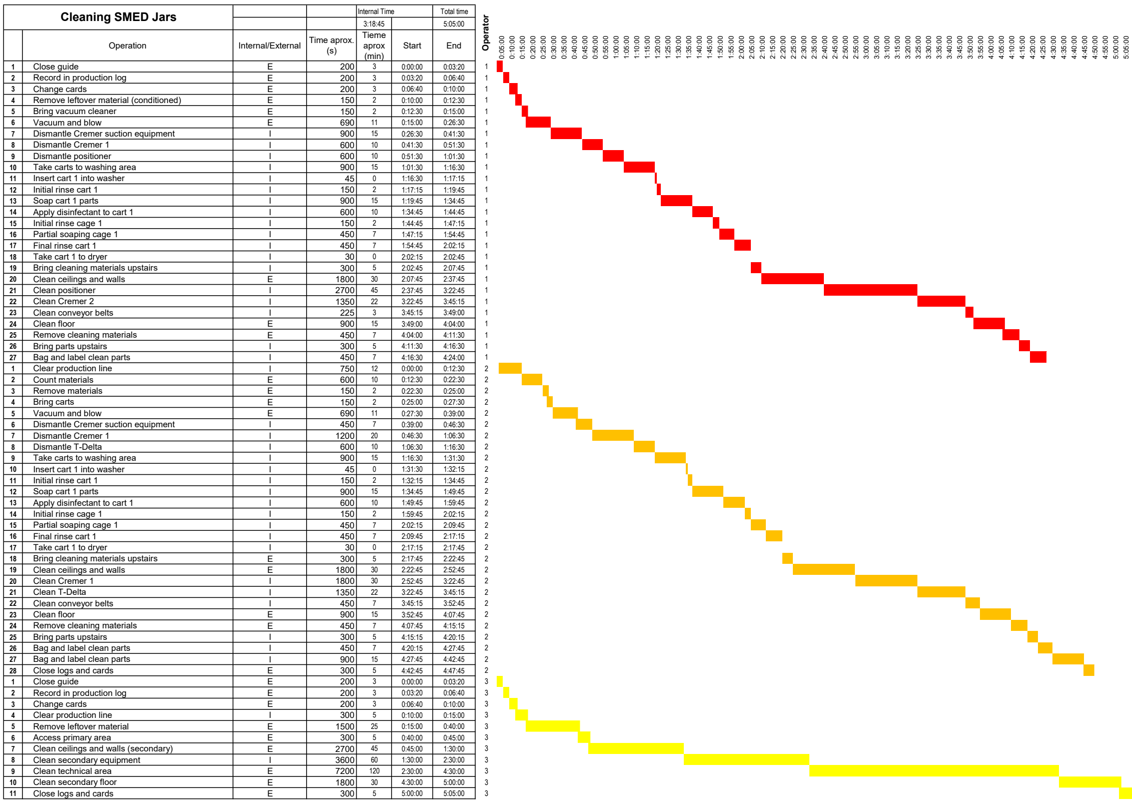

Figure 11: Gantt Diagram of Cleaning Procedure

Tasks	Time (s)	Frequency (1/s)	MUDA/VA	Operator
Change dispenser box	50	0.001	VA	OP 1
Change case box	50	0.0004	VA	OP 2
Change leaflet box	50	0.0004	VA	LINE COORDINATOR
Change label	70	0.0005	VA	LINE COORDINATOR
Load cases	20	0.004	VA	LINE COORDINATOR
Load leaflets	40	0.004	VA	LINE COORDINATOR
Check machine parameters	10	0.0002	MUDA	LINE COORDINATOR
Case quality control	20	0.0022	VA	LINE COORDINATOR
Record quality control result	45	0.0022	MUDA	LINE COORDINATOR
Monitor rejection 2	10	0.02	VA	OP 2
Discard boxes	84	0.0012	VA	OP 2
Perform label rejection control (Rejection 1)	20	0.01	VA	LINE COORDINATOR
Pallet shipment	180	0.00078	VA	OP 2
Push bottles onto the cartoner conveyor	30	0.001	MUDA	OP 2
Place dispenser in case	0.75	1	VA	OP 1
Label box	1.5	0.2	VA	OP 2
Place cases into boxes (5 at a time)	1	0.2	VA	OP 2
Palletize	1	0.2	VA	OP 2
Make box	2	0.2	VA	OP 2

Table 6: Line Tasks Before Improvement

CHECK-LIST DE LIMPIEZA

Linea: BOTES PRIMARIO

OPERARIO 1

Turno:

Fecha:

Nº	TAREA	NOMBRE OPERARIO/A	HORA INICIO	HORA FIN	TIEMPO REAL
1	CERRAR GUIA				3
2	ANOTAR EN DIARIO DE PRODUCCIÓN				3
3	TARJETAS CAMBIO COLOR				3
4	SACAR MATERIAL SOBRANTE(ACO)				3
5	TRAER ASPIRADORES,Y ASPIRAR(2 PERS)				23
6	DESMONTAR EQUIPO SUCCION CREMER (2PERS)				15
7	DESMONTAR CREMER 2 (OPCIONAL)				
8	DESMONTAR POSICIONADORA				10
9	SOPLAR				10
10	BAJAR CARROS, JAULAS AL LAVADERO				15
11	LIMPIEZA PIEZAS LAVADERO (2PERS)				5
12	PASAR AGUA PIEZAS				5
13	PASAR JABON PIEZAS				30
14	ACLARAR PIEZAS				15
15	ECHAR DESINFECTANTE (20MIN)				20
16	ACLARAR PIEZAS (3MIN)				5
17	DEJAR PIEZAS EN EL SECADERO				5
18	SUBIR MATERIAL LIMPIEZA				15
19	LIMPIEZA TECHO Y PAREDES(2PERS)				30
20	LIMPIEZA POSICIONADORA				45
21	LIMPIEZA CREMER 2 (OPCIONAL)				
22	LIMPIEZA PIEZAS TRANSPORTADORA				15
23	LIMPIEZA ESCALERAS MESA MATERIAL ADICIONAL				7
24	LIMPIAR SUELO(2 PERS)				30
25	BAJAR MATERIAL DE LIMPIEZA AL LAVADERO				5
26	SUBIR PIEZAS SECAS(EMBOLSAR Y ETIQUETAR PIEZAS LIMPIAS)(2 PERS)				15

Figure 12: Standard Procedure Checklist Example